

INVENTORY MANAGEMENT SYSTEM

Data Analyst Project Report

Excel • MySQL • SQL • Power BI • DAX

*A practical inventory analysis project covering data preparation,
SQL-based analysis and interactive business reporting.*

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PROJECT OVERVIEW

The **Grocery Inventory Management** project is a data analytics project designed to analyze and monitor grocery inventory data. The project uses **Excel, MySQL, Power BI, and DAX** to clean, manage, analyze, and visualize the data.

The dataset contains **990 records and 16 fields**, including product details, supplier information, stock quantity, sales volume, reorder levels, expiration dates, unit prices, and warehouse locations.

The project follows an end-to-end data analytics process:

Excel → MySQL → Power BI → DAX → Interactive Dashboard

The analysis focuses on important inventory metrics such as total stock quantity, total sales, total products, expired products, products requiring restocking, and total inventory value. The Power BI dashboard also provides visual analysis based on product category, warehouse, supplier, product status, and reorder quantity.

The project demonstrates how raw inventory data can be transformed into **meaningful insights and interactive visualizations** that help in monitoring inventory and understanding overall inventory performance.

PROBLEM STATEMENT

Managing grocery inventory requires accurate information about **stock levels, sales, product availability, expiration dates, reorder requirements, suppliers, warehouses, and inventory value**.

The raw inventory data contains a large amount of information, making it difficult to identify important inventory conditions through manual analysis. There is a need to organize and analyze the data systematically to understand inventory performance and identify products that require attention.

This project addresses the following problems:

- Difficulty in monitoring overall **stock quantity and sales**.
- Identifying products that have reached or fallen below their **reorder level**.
- Monitoring **expired products**.
- Understanding inventory distribution across **categories and warehouses**.
- Analyzing the number of products handled by different **suppliers**.
- Understanding **reorder quantities** across categories.
- Calculating and monitoring the **total inventory value**.
- Presenting the analysis in a clear and interactive format for easier understanding.

The project therefore aims to transform the raw grocery inventory data into **structured analysis and meaningful visual insights** using **Excel, MySQL, SQL, Power BI, and DAX**.

OBJECTIVES

The main objective of the **Grocery Inventory Management** project is to analyze grocery inventory data and develop a clear, interactive dashboard that provides useful information about stock, sales, products, suppliers, and inventory value.

The project aims to:

- **Clean and prepare** the raw inventory data using Excel.
- **Store and manage** the cleaned data using MySQL.
- **Perform SQL analysis** to understand inventory-related information.
- Analyze **stock quantity and sales** across different product categories.
- Identify **products that require restocking** based on their stock and reorder levels.
- Monitor **expired products** using expiration date information.
- Analyze products based on **suppliers and warehouse locations**.
- Calculate the **total inventory value** using stock quantity and unit price.
- Create required **DAX calculations** for important inventory metrics.
- Develop an **interactive Power BI dashboard** using KPI cards and charts.
- Present the analyzed data in a simple visual format to make **inventory information easier to understand**.

Overall, the project aims to transform raw grocery inventory data into **structured analysis, visual insights, and an interactive dashboard** using Excel, MySQL, Power BI, and DAX.

DATASET DESCRIPTION

The **Grocery Inventory Management** dataset contains information related to grocery products, suppliers, inventory levels, sales, warehouses, and product expiration. The dataset consists of **990 records and 16 columns**.

The dataset includes the following important fields:

Field	Description
Product_ID	Unique identification number of each product
Supplier_ID	Identification number of the supplier
Product_Name	Name of the product
Supplier_Name	Name of the supplier
Category	Category of the product
Warehouse_Location	Location where the product is stored

Field	Description
Product_Status	Current status of the product
Date_Received	Date when the product was received
Last_Orderdate	Date of the last order
Expiration_Date	Expiration date of the product
Stock_Quantity	Quantity of products currently available
Reorder_Level	Stock level at which reordering is required
Reorder_Quantity	Quantity recommended for reorder
Unit_price	Price of one unit of the product
Sales_Volume	Number/volume of products sold
Inventory_Turnover_Rate	Rate at which inventory is sold and replaced

The dataset was first prepared in **Excel**, then used for **MySQL/SQL analysis** and **Power BI visualization**. These fields were used to calculate key inventory metrics and create the final **Inventory Management Dashboard**.

DATA PREPARATION–EXCEL

The raw grocery inventory dataset was initially prepared and cleaned using **Microsoft Excel** before importing it into MySQL. The purpose of data cleaning was to improve data quality and ensure that the dataset was properly structured for further analysis.

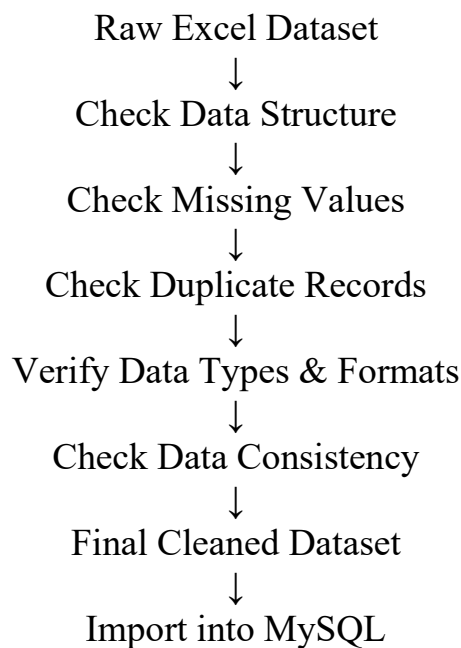
The following activities were performed:

- Checked the dataset for **missing or blank values and duplicate records**.
- Verified the **column names and data structure**.
- Checked and maintained appropriate **data formats**, especially for date and numerical fields.
- Reviewed important fields such as **Stock Quantity, Reorder Level, Reorder Quantity, Unit Price, and Sales Volume**.

	A	B	C	D	E	F	G	H	I
	Product_Name	Category	Supplier_Name	Warehouse_Location	Product_Status	Product_ID	Supplier_ID	Date_Received	Location
2	Bell Pepper	Fruits & Vegetables	Eimbee	20 Pennsylvania Parkway	Discontinued	290176255	433482450	03/01/2024	
3	Vegetable Oil	Oils & Fats	Digitube	03643 Oakridge Lane	Backordered	795698856	48547165	04/01/2024	
4	Parmesan Cheese	Dairy	BlogXS	73 Graedel Street	Discontinued	281462641	829950739	04/01/2024	
5	Carrot	Fruits & Vegetables	Avaveo	44801 Myrtle Center	Discontinued	115819869	228673079	05/01/2024	
6	Garlic	Fruits & Vegetables	Katz	6195 Monterey Center	Discontinued	132024809	242817685	05/01/2024	
7	Lemon	Fruits & Vegetables	Yata	5141 Anniversary Crossing	Backordered	701452550	758494524	05/01/2024	
8	Coconut Sugar	Grains & Pulses	Lazz	38583 2nd Pass	Active	106268536	232743305	05/01/2024	
9	Anchovies	Seafood	Zoonder	86 Porter Junction	Discontinued	428799478	9000119	06/01/2024	
10	Cheese	Dairy	Oozz	05518 Saint Paul Street	Active	823805378	963533049	06/01/2024	
11	Yogurt	Dairy	Jaxnation	14042 Dottie Avenue	Backordered	232658144	315241628	06/01/2024	
12	Cheddar Cheese	Dairy	Gabcube	5 Oxford Pass	Active	623939939	938779384	06/01/2024	
13	Avocado Oil	Oils & Fats	Dabtype	3622 Anhalt Lane	Backordered	903439640	549199424	07/01/2024	
14	Orange	Fruits & Vegetables	Flashdog	3 Bashford Court	Backordered	368995324	26563112	08/01/2024	
15	Digestive Biscuit	Bakery	Devshare	2 Carioca Crossing	Backordered	415383129	915899007	09/01/2024	
16	Cauliflower	Fruits & Vegetables	Cogilith	01 Northfield Road	Active	478438207	558389088	09/01/2024	
17	Pear	Fruits & Vegetables	Roodel	2004 Cody Plaza	Discontinued	940712261	257062146	09/01/2024	
18	Egg (Turkey)	Dairy	Bluezoom	94862 Beifuss Terrace	Backordered	129983882	61519076	10/01/2024	
19	Ricotta Cheese	Dairy	Skimia	10342 Division Place	Active	361274273	57196858	10/01/2024	

- Ensured that product, supplier, category, and warehouse information was consistently organized.
- Prepared the cleaned dataset for **import into MySQL**

Data Cleaning Process

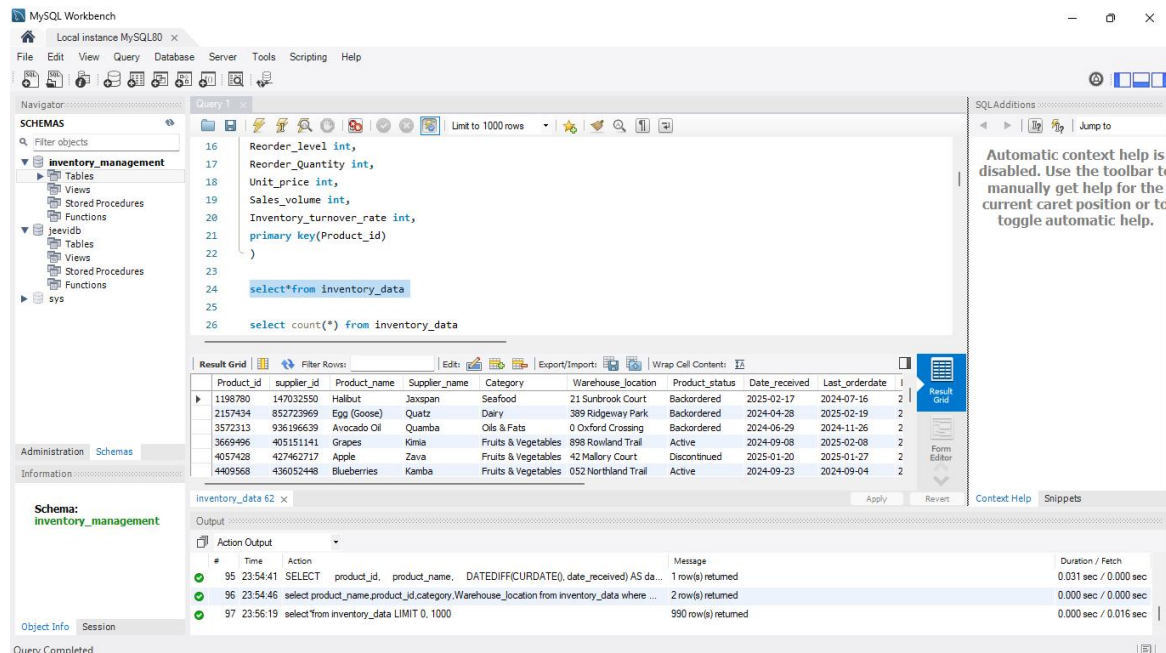


MYSQL DATABASE & SQL ANALYSIS

After completing the data cleaning process in Excel, the cleaned grocery inventory dataset was imported into **MySQL** for structured storage and analysis. A database named `inventory_management` was created, and the inventory data was stored in the `inventory_data` table.

The Product_ID was used as the primary key to uniquely identify each product. The table contains important information related to products, suppliers, categories, warehouses, stock, sales, reorder levels, pricing, and expiration dates.

SQL queries were then written to explore and analyze the inventory data. The analysis included checking the total number of records, identifying distinct products and categories, checking suppliers and warehouse locations, and verifying missing values.



Further SQL analysis was performed to understand stock availability, sales volume, product status, expiration information, and reorder requirements. Queries were also used to identify products that needed restocking by comparing Stock Quantity with Reorder Level.

POWER BI DATA MODEL

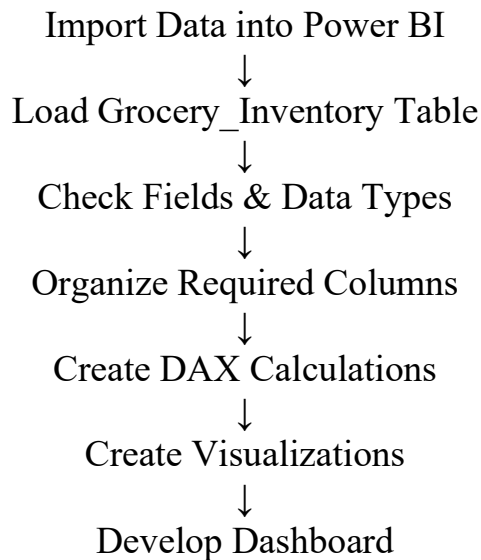
After completing the SQL analysis, the inventory data was imported into **Microsoft Power BI** for visualization and dashboard development. The Power BI model contains a single table named Grocery_Inventory, which includes all the required product, supplier, stock, sales, warehouse, and inventory information.

Since the project uses a single table, there was no requirement to create relationships between multiple tables. The fields in the Grocery_Inventory table were organized and assigned appropriate data types so they could be used correctly in visualizations and calculations.

The data model contains important fields such as Product_ID, Product_Name, Category, Supplier_Name, Warehouse_Location, Product_Status, Stock_Quantity, Reorder_Level, Reorder_Quantity, Unit_price, Sales_Volume, and Expiration_Date.

The prepared data model was then used to create **DAX calculations, KPI cards, charts, and the final Inventory Management Dashboard.**

Power BI Data Model Process



DAX CALCULATIONS

After creating the Power BI data model, **DAX (Data Analysis Expressions)** was used to create calculations required for the Inventory Management Dashboard. These calculations helped convert the raw inventory fields into meaningful **KPIs and analytical values**.

The following DAX calculations were used in the project:

- 1. Total Stock Quantity** -Calculates the total quantity of products available in the inventory.
- 2. Total Sales**-Calculates the total sales volume from all products.
- 3. Total Products**-Calculates the total number of products in the dataset.
- 4. Products Need Restock**-Identifies products where the stock quantity is less than or equal to the reorder level.
- 5. Total Inventory Value**-Calculates the total value of the available inventory using stock quantity and unit price.
- 6. Expired Products**-The expiration date was analyzed to identify products whose expiration date had passed.

These DAX calculations were then used in **KPI cards and Power BI visualizations** to create the final Inventory Management Dashboard.

DASHBOARD AND VISUALIZATION

After completing the data analysis and DAX calculations, an interactive **Inventory Management Dashboard** was developed using **Power BI**. The dashboard was designed to present important inventory information in a simple and visually understandable format.

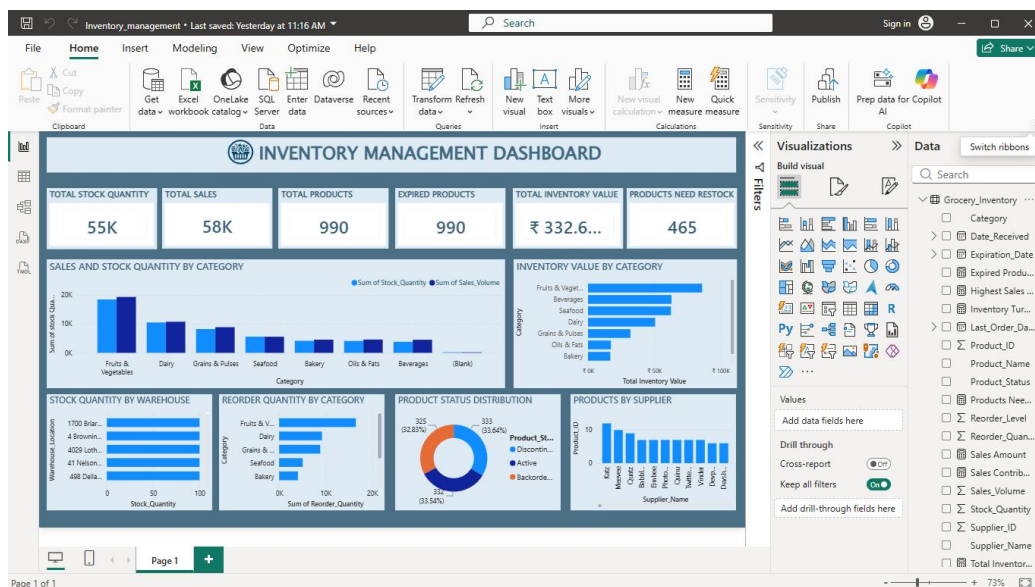
The dashboard contains **KPI cards** and different charts to provide an overall view of inventory performance.

Key KPI Cards

- **Total Stock Quantity** – 55K
- **Total Sales** – 58K
- **Total Products** – 990
- **Expired Products** – 990
- **Products Need Restock** – 465
- **Total Inventory Value** – ₹332.65471K

Dashboard Visualizations

1. **Stock Quantity & Sales by Category**
Compares stock quantity and sales across different product categories.
2. **Inventory Value by Category**
Shows the total inventory value for each product category.



3. **Stock Quantity by Warehouse**
Displays the distribution of available stock across warehouse locations.
4. **Product Status Distribution**
Shows the distribution of products based on their current status.
5. **Products by Supplier**
Represents the number of products associated with each supplier.
6. **Reorder Quantity by Category**
Shows the reorder quantity required across different product categories.

The combination of **KPI cards** and **visualizations** provides a consolidated view of the grocery inventory data and makes it easier to understand important inventory metrics.

KEY INSIGHT

- The dataset contains **990 products** across multiple categories and suppliers.
- Total stock quantity is approximately **55K units**.
- Total sales volume is approximately **58K units**.
- **465 products** require restocking based on stock quantity and reorder level.
- Inventory is distributed across different warehouses and categories.
- Supplier analysis shows how products are distributed among suppliers.
- Reorder quantity varies across product categories.
- Total inventory value is approximately **₹332.65K**.
- Expiration-date analysis helps identify products requiring attention.
- SQL analysis and Power BI visualizations make inventory patterns easier to understand.

Overall: The project provides a structured view of inventory levels, sales, product distribution, suppliers, warehouses, expiry status, restocking requirements, and inventory value, making the data easier to understand and monitor for inventory management purposes.

BUSINESS USE CASES

Business Area	Use Case	How the Project Helps
Inventory Monitoring	Track stock levels	Monitor available stock across products and categories
Restocking	Identify products needing restock	Identify products where stock is at or below the reorder level
Sales Analysis	Monitor sales volume	Analyze sales across different product categories
Expiry Management	Track expired products	Identify products requiring attention based on expiration dates
Inventory Valuation	Calculate inventory value	Estimate the value of available inventory using stock quantity and unit price
Warehouse Management	Monitor warehouse stock	Understand how inventory is distributed across warehouse locations

Business Area	Use Case	How the Project Helps
Supplier Management	Analyze suppliers	Monitor the number of products associated with different suppliers
Category Analysis	Compare product categories	Analyze stock, sales, inventory value, and reorder quantities by category
Inventory Planning	Support purchasing decisions	Use stock and reorder information for replenishment planning
Management Reporting	Dashboard-based reporting	Provide managers with a visual summary of important inventory metrics

END-TO-END PROJECT WORKFLOW

Step	Workflow Stage	Activities Performed	Tool Used
1	Data Collection	Collected and reviewed the raw grocery inventory dataset	Excel
2	Data Cleaning	Checked missing values, duplicates, formats, and data consistency	Excel
3	Data Preparation	Prepared the final cleaned dataset for analysis	Excel
4	Database Creation	Created the inventory database and table	MySQL
5	Data Import	Imported the cleaned dataset into MySQL	MySQL
6	SQL Analysis	Analyzed products, stock, sales, suppliers, warehouses, expiry, and restocking requirements	SQL
7	Power BI Import	Imported the inventory data into Power BI	Power BI
8	Data Model Preparation	Checked the table structure, fields, and data types	Power BI
9	DAX Calculations	Created KPI calculations such as total stock, sales, restocking, and inventory value	DAX

Step	Workflow Stage	Activities Performed	Tool Used
10	Visualization	Created KPI cards, charts, and other visualizations	Power BI
11	Dashboard Development	Designed and formatted the Inventory Management Dashboard	Power BI
12	Insights & Analysis	Interpreted the results and identified important inventory patterns	SQL / Power BI

SKILLS DEMONSTRATED

- **Excel** – Data cleaning & preparation
- **MySQL & SQL** – Database management & data analysis
- **Power BI** – Dashboard & visualization
- **DAX** – KPI calculations
- **Data Analysis** – Inventory insights & business understanding
- **Data Visualization** – Charts, KPIs & reporting
- **Problem Solving** – Inventory-related analysis

CONCLUSION

The Grocery Inventory Management project successfully transformed raw inventory data into meaningful and structured information. The data was cleaned and prepared using Excel, stored and analyzed using MySQL and SQL, and then transformed into an interactive Power BI dashboard using DAX calculations.

The project provides insights into stock levels, sales, products, suppliers, warehouses, expiry status, restocking requirements, and inventory value. The final dashboard presents these insights in a clear and visual format, making the inventory data easier to understand and monitor.

Overall, this project demonstrates the practical application of Excel, MySQL, SQL, DAX, Power BI, data analysis, and data visualization in solving a real-world inventory management problem.

